

1 Light-sharing control for tracker-based agrivoltaics
2 under a yield floor: a simulation study of what
3 learning adds and what shade-response uncertainty
4 costs

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7 **Abstract**

Single-axis trackers let agrivoltaic arrays shift their daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a yield-retention floor of 90% with a differentiable simulator of the whole system. On 92 held-out crop–site pairs and the proportional-growth shade response, light-sharing rules allowed arrays 2.1–2.9 times as dense as static arrays at equal mean retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24, with the floor met in 69–76% of seasons (static arrays: 93%). The crop response to shade is the least certain input: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, so we turned the learned curves and field trials into ensembles of plausible responses by gradient descent through the simulator. Within these ensembles, a density chosen under the mean response left about 70% of pairs at the floor when another plausible response held, against 85–87% for an oracle design with the true response; a floor raised by 0.02 and an ensemble-quantile rule gave no detectable difference at equal compliance, and neither was safe outside the ensemble. Pooled over three splits (76 pairs), a neural controller initialised from the written phenology rule beat it by 0.044 LER but a CMA-ES-optimised window by only 0.006 (95% interval 0.001–0.012, one training seed); training on ensembles or a biomass signal did not help within our budget. Under illustrative costs, light sharing breaks even with the open field at 70–92 USD MWh^{−1} (static arrays at 59) and a conventional plant is more profitable above 48. For maize and soybean (53%

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of the upscaled 14 PWh yr^{-1}) the simulator’s bias is uncertain.

8 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
9 Crop modelling, Differentiable simulation, Decision under uncertainty

10 1. Introduction

11 Two of the defining challenges of this century compete for the same re-
12 source: land. The energy transition requires terawatts of new solar pho-
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
14 to two hectares per megawatt. Food production must rise while crops are
15 exposed to more frequent heat waves and droughts. Warming alone is pro-
16 jected to reduce global yields of maize, wheat, rice and soybean by several
17 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation
18 of climate and crop models shows these impacts emerging earlier than pre-
19 viously expected (Jägermeyr et al., 2021). Croplands are also where solar
20 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-
21 sible and close to rural electricity demand, so converting farmland into solar
22 farms is tempting and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
26 a hectare can deliver more food plus energy than the two uses side by side,
27 with land-equivalent ratios (LER) well above one in field trials and mod-
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
30 climate. It lowers canopy temperature and evaporative demand, conserves
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-
32 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-
34 didate adaptation measure against heat and drought stress, as reviewed by
35 Mamun et al. (2022), in addition to a mitigation measure through renew-
36 able electricity. The same shade, however, also reduces the light available for
37 photosynthesis. Regulators increasingly require AV systems to retain most
38 of the reference crop yield. The German pre-standard DIN SPEC 91434, for
39 example, requires at least 66% of the reference yield (DIN, 2021), and other
40 schemes and many farmers set stricter floors. How much electricity can be
41 produced without sacrificing food security depends on the crop, the local

42 climate, the year and the growth stage.

43 Most AV designs fix the geometry of the array, i.e. the ground coverage
44 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
45 generic rules of thumb. Single-axis trackers add a degree of freedom: the
46 modules can be rotated away from the sun to let light through when the crop
47 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
48 sun to maximise electricity. Choosing the rotation every day is a sequential
49 decision problem. The value of light depends on the crop’s phenological stage,
50 soil water and upcoming weather, and the effect of a decision appears in the
51 harvest months later. Recent work in this journal has used evolutionary
52 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
53 2026). Machine learning for crop management has mainly used reinforcement
54 learning with black-box simulators (Gautron et al., 2022), which would need
55 very many simulated seasons for global, multi-decadal assessments.

56 Two methodological strands are relevant. Differentiable process models,
57 which couple physical simulators with learning, allow parameters and poli-
58 cies to be fitted by gradient descent through the simulator (Reichstein et al.,
59 2019; Shen et al., 2023), and we use them for both calibration and control.
60 Decision making under deep uncertainty asks for designs that perform accept-
61 ably across plausible models and not only under the best estimate (Marchau
62 et al., 2019), and scenario-based designs have finite-sample guarantees un-
63 der convexity assumptions that we do not meet (Calafiore and Campi, 2006;
64 Campi and Garatti, 2008); our design rules are heuristic counterparts and
65 we compare them with a simple safety margin.

66 Two questions follow. First, how much of the electricity that an agri-
67 voltaic system can produce under a yield-retention floor comes from the array
68 density, and how much from dynamic light-sharing control of the trackers?
69 Second, how robust is that answer to the weakest part of any such study, the
70 crop response to shade, and does a controller learned from a simulator add
71 anything to a well-designed hand-written rule? Answering them requires
72 a simulator that couples the physics of light, PV, microclimate and crop
73 growth, a protocol in which all controllers face the same design constraints,
74 a check of the shade response against field data, and enough sites and seasons
75 to separate real effects from noise.

76 This paper makes three contributions. The first is a quantification of
77 light-sharing control under a yield floor. Light sharing is parameterised by
78 one daily level between full backtracking and the rotation that minimises the
79 shadow on the crop, and we compare static arrays, a seasonal schedule, a rule

that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever on electricity at a given floor, although economically it pays only at high prices (item 6). At the same mean yield retention of 90%, arrays with light-sharing control carry 2.1 to 2.9 times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but milder than the meta-analysis for maize and soybean, whereas one maize trial

is milder still; the gains for those two crops are therefore uncertain and possibly overestimated (Section 3.7).

4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule but only 0.006 over a window optimised on the same training seasons. Training it on ensembles of shade responses, or giving it a farm-measurable biomass signal, does not help, and with the wider ensembles it did not retain its margin over the written rule in our training budget (Section 3.8).
5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 8 points for 0.03 LER, and a floor raised by 0.02 under the mean response gives no detectable difference at equal compliance (the intervals are too wide to establish equivalence); no rule is safe outside the ensemble, so the ensemble is useful to size a margin (Section 3.8).
6. Under illustrative costs, light sharing breaks even with the open field at 70 to 92 USD MWh⁻¹ (static arrays at 59), learning is worth about 0.4–0.5 kUSD ha⁻¹ yr⁻¹ over an optimised window at 90 USD MWh⁻¹, and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh⁻¹ (Section 3.9).
7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr⁻¹, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.11).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the

GGCMI Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below -8°C , in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels $+1.5$, $+2$ and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

Hourly irradiance. Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

Array geometry and PV electricity. We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The rotation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the rows when the modules are horizontal, which is the stow position during field operations. Let ψ be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,

2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ otherwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines three terms: beam irradiance on the module, reduced by the fraction of the module shaded by the neighbouring row, $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A conventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the energy reference E_{PV} .

Validation of the PV model. We compared the simulated specific yield of an isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2 (European Commission, Joint Research Centre, 2022), for a horizontal north-south single-axis tracker with 14% losses, at 30 randomly chosen sites (29 returned data). Across sites, the simulated and PVGIS specific yields were correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARA2, ERA5 and NSRDB). The benchmark covers an isolated row at low density only; electricity at high density, with row-to-row shading and diffuse-light effects, was not validated.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop's shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

221 κ is uncertain. Field studies under PV modules report small reductions
 222 of daytime crop or canopy temperature and little change in air temperature
 223 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 224 calibrated on air temperature do not transfer directly to canopy temperature
 225 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 226 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 227 for all methods (Section 3.2). Nocturnal warming under the modules is not
 228 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 229 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 230 from the reduced radiation and canopy temperature. Soil water conserva-
 231 tion under the modules (Hassanpour Adeh et al., 2018; Elamri et al., 2018)
 232 therefore emerges from the model, but rain redistribution by the modules
 233 (Marrou et al., 2013a) is ignored.

234 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 235 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 236 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

237 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
 238 follows.

- 239 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 240 only on thermal time, with senescence accelerated by heat and drought
 241 stress. Shade therefore reduces the daily biomass increment but not
 242 leaf-area development.
- 243 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 244 treme temperature T_{ext} of the canopy maximum temperature.
- 245 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
 246 drought (Woli et al., 2012) is computed from a daily soil-water balance
 247 with curve-number runoff, drainage and root water uptake. Irrigated
 248 fields have $f_{\text{Water}} = 1$.

249 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 250 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 251 drade et al., 1999). It also ignores the differences in shade tolerance be-
 252 tween crop types. In the meta-analysis of Laub et al. (2022), maize and

253 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 254 reals and tuber crops initially lost less than proportionally; shading trials
 255 also show marked effects of shade on maize development (Ramos-Fuentes
 256 et al., 2023). We added a stage-dependent penalty. During a critical window
 257 ($0.45 \leq TT/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 258 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 259 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 260 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 261 0.5, for which no specific curve was available) and lowest for potato (0.2).
 262 These values are assumptions that reproduce this ranking; they are not cal-
 263 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 264 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 265 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 266 makes shading around flowering more damaging than proportional. The net
 267 effect is closer to the field evidence than either assumption alone.

268 The thermal-time requirement to maturity was calibrated for every site as
 269 the mean thermal time between the GGCMI sowing and maturity dates under
 270 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 271 The cultivar is thus adapted to the present climate, and warming shortens the
 272 season. Growth stops smoothly once the thermal-time requirement is met.
 273 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 274 is represented by a rainfed and an irrigated sub-field in proportion to its
 275 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 276 *and* electricity are area-weighted over the two sub-fields.

277 2.3. SALS: simulation-trained adaptive light sharing

278 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 279 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 280 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 281 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 282 are the information available on day t :

- 283 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 284 terception f_{Solar} , relative soil water and the previous day’s ARID;
- 285 • a *forecast* of the day’s maximum temperature (relative to T_{heat} and
 286 T_{opt}), irradiance and ET_0 ;

287 • static information: the irrigation status of the sub-field, the array’s
 288 GCR g , latitude, day of season and the crop.

289 Forecasts are perturbed with random errors, independently every day: Gaus-
 290 sian with a standard deviation of 2 °C for maximum temperature, and mul-
 291 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 292 The physics always uses the true weather. The same forecast errors are
 293 applied to the rule-based controllers. The controller is conditioned on g .
 294 During training, each season is simulated with a GCR close to the density
 295 that is sustainable at its training site. This is the largest value on a coarse
 296 grid (0.05, 0.10, ..., 0.50) at which seasonal sharing keeps the mean retention
 297 of the site’s training seasons at or above ρ , plus a uniform perturbation in
 298 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 299 trol the densities that design selection will later pick. Outside the cropping
 300 season the trackers backtrack for maximum electricity.

301 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 302 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 303 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

304 which combines an exact (hinge) and a quadratic penalty for the yield-
 305 retention floor ρ . Because r_i and e_i are outputs of the differentiable simulator,
 306 the exact gradient of Eq. (3) is obtained by back-propagation through the 365
 307 daily steps of every season (back-propagation through time). This is the key
 308 difference from reinforcement learning with black-box simulators (Gautron
 309 et al., 2022), which estimates gradients from sampled returns.

310 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
 311 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
 312 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
 313 and evaluation. The network is fine-tuned through the simulator with Adam
 314 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
 315 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
 316 a numerically stable form, so that gradients remain finite even under extreme
 317 heat.

318 *Initialisation.* Trained from random weights, the controller converged to
 319 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
 320 the seasonal-sharing rule, because the gradient of electricity with respect to u

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{PV,i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

321 is small and of the same sign for most days of a season. We therefore initialise
322 the network by behaviour cloning of the phenology rule of Section 2.4: binary
323 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
324 1.7 million day-states visited by the rule in the training seasons, with 1500
325 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
326 policy away from this initialisation wherever doing so improves the objective.
327 Consequently, the gain of SALS over the phenology rule measures what fine-
328 tuning through the simulator adds to that rule, not what can be learned
329 without it.

330 *2.4. Deployable design selection and baselines*

331 A real array has one fixed density, which must be chosen before it is built.
332 Every method, SALS included, therefore uses the same *history-based design*
333 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
334 for which the method’s mean yield retention over the site’s 2001–2014 sea-
335 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
336 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
337 at least 80% of the historical seasons. The historical seasons of the test sites
338 are never used for training, and the weather of the evaluation years is never
339 used for design. We compare four controllers:

- 340 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 341 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
342 cropping season;
- 343 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
344 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
345 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
346 track the sun and shade the crop;
- 347 • *SALS*: the learned controller.

348 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
349 are the references. Because the design rule is identical for all controllers, dif-
350 ferences between methods reflect the quality of control and the array density
351 that each controller can sustain under the yield-retention floor. We also com-
352 pare the controllers at identical GCR over the whole grid (matched-density
353 frontiers).

354 2.5. A shade-response model learned from field data

355 The simulator needs a crop response to shade that is supported by ob-
356 servations. We learned one from the open data of the meta-analysis of Laub
357 et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observa-
358 tions of yield relative to an unshaded control at relative shade reductions of
359 0–90% from 58 studies, with crop and crop type, experiment type (artificial
360 shading or intercropping), climate zone, irradiation and shade type. Rela-
361 tive yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$;
362 maximum a posteriori estimation, so this is a hierarchical shrinkage regres-
363 sion with bootstrap intervals and not a full Bayesian model). We compared
364 six models: proportional loss $y = 1 - s$; one curve for all crops; one curve
365 per crop type without pooling; one curve per crop type with partial pooling
366 (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5);
367 partial pooling with covariates on $\log \beta$ (standardised irradiation, subtrop-
368 ical zone, solar-panel shade and intercropping); and gradient-boosted trees
369 on the same inputs. Fits use L-BFGS. Models were compared by leave-one-
370 study-out cross-validation, in which every study is held out once, which tests
371 the prediction of a new study and not of a new observation in a known study.
372 Intervals for the crop-type curves and covariate effects come from a cluster
373 bootstrap over studies (150 resamples).

374 2.6. Shade-response uncertainty: gradient calibration, robust controllers and 375 design

376 The crop’s response to shade is the least certain part of the simulator,
377 so we treat it as an unknown with a distribution. Two parameters per crop
378 enter the simulator: a radiation-use compensation c (growth is multiplied by
379 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
380 and a multiplier h on the flowering-window harvest-index penalty. Because
381 the simulator is differentiable, both can be fitted by gradient descent through
382 whole simulated seasons to published yield–shading observations. The ob-
383 servations are the relative yields at 20% and 40% shading of the data-driven
384 curves of Section 2.5 for the five crops (median, with an error equal to the
385 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice
386 takes the curve of C3 cereals, which contains no rice), the field trials on rice,
387 wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-
388 Fuentes et al., 2023) (each season of each trial as a separate observation, error
389 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes
390 et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses
391 the same procedure with a target error of 0.2 for the learned curves, which
392 is the between-study spread of a new study (Section 3.7); the spread of
393 the resulting members is smaller (standard deviation of relative yield across
394 members 0.04–0.09 at 20% shading and 0.05–0.12 at 40% shading, against
395 0.02–0.04 and 0.03–0.06 for E2), because the fit also weights the field trials
396 and the priors. An earlier ensemble, E1, was fitted in the same way to curves
397 read by hand from the figures of Laub et al. (2022) and was used to train
398 the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and
399 $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h)
400 after every step, which is why a response milder than the data cannot be
401 lowered further. We fitted 12 ensemble members by randomised maximum a
402 posteriori estimation: every member sees the observations perturbed by their
403 errors and a prior draw perturbed by its width, and is optimised with Adam
404 (40 steps, with the loss still falling slightly at the last step; 40 training-block
405 seasons per crop and member; shading of the static arrays cached). The
406 ensemble is a device for generating plausible responses and not a validated
407 posterior: it has no coverage check, and for some parameters (the harvest-
408 index multiplier of wheat and potato) its spread equals that of the prior.
409 The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12),
410 chosen in advance. The ensemble is used in three ways. All training and
411 evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes

412 and the calibration of one ensemble about 7 minutes.

413 *Tuned rules.* The written phenology rule shares light while thermal time
414 lies between 0.05 and 1.0 of the season requirement. To avoid comparing a
415 trained controller with an untuned rule, we tuned this window on training-
416 block seasons of split 0 (500 random seasons, baseline climate, main shade
417 response) with the objective used for all rules: the relative electricity at the
418 density where the mean retention equals the floor, found by interpolation over
419 five densities (0.1 to 0.5). A 4×3 grid of start (0, 0.05, 0.15, 0.3) and end
420 (0.6, 0.8, 1.0) gave the grid-tuned window. A five-parameter schedule (start
421 in $[0, 0.6]$, end in $[0.5, 1.2]$, two ramp widths and a level in $[0, 1]$, all through
422 logistic or exponential transforms) was then optimised by CMA-ES (Hansen
423 and Ostermeier, 2001) with 300 seasons, population 8 and 25 generations
424 (200 evaluations) per run; run 1 started at the grid-tuned window and runs 2
425 and 3 at random points, and the run with the best training objective (run 2)
426 was evaluated. Run 3 stopped in a worse optimum (objective 0.345 against
427 0.377), so three runs give no guarantee of global optimality.

428 *Robust and adaptive controllers.* SALS-robust has the architecture and
429 training of SALS (Section 2.3), but every training season is simulated with the
430 response of a randomly drawn ensemble member that the controller does not
431 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
432 densities differ between members. SALS-adaptive additionally observes the
433 realised biomass of the crop under the modules relative to an open reference
434 plot, a quantity that can be measured on the farm by remote sensing or by
435 a reference plot. A feedback rule that shares light whenever this ratio falls
436 below a target serves as the hand-written counterpart; its target (1.0) was
437 chosen on training-block seasons under the ensemble.

438 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
439 largest GCR whose historical retention meets the floor. Under shade-response
440 uncertainty we choose the largest GCR that meets the floor in at least 75%
441 of five ensemble members, and compare it with a design under the ensemble
442 mean and with an oracle design under the true response. Evaluation of this
443 rule is leave-one-out: the truth is the held-out member and the design uses
444 the other four.

445 *Design rules compared.* With the ensemble members indexed k and one
446 member held out as the truth, the density of a pair is chosen from its 2001–
447 2014 seasons as (i) the *oracle*: the largest GCR whose mean retention under
448 the truth meets the floor; (ii) the *mean-response design*: the same rule under
449 the mean response of the other four members (the mean of their parameters);

450 (iii) a *margin*: the same rule under that mean response with the floor raised
 451 to 0.92, 0.94 or 0.96; and (iv) the *ensemble rule*: the largest GCR at which
 452 the mean retention under at least a share q of the other four members meets
 453 the floor ($q = 0.5$: two of four; 0.75 : three of four; all: four of four). Out
 454 of the ensemble, the five members and their mean (the mean of all twelve
 455 members for the mean-response design) are used in the same way and the
 456 truth is one of the four fixed responses below. The margin grid was chosen
 457 after a first analysis of one ensemble and is not pre-specified.

458 *Stress tests.* Controllers and rules are also evaluated under four fixed
 459 responses, the proportional response, the response calibrated to the meta-
 460 analysis, the response fitted to the field trials and a harsh response (propor-
 461 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 462 and harsh responses lie outside the range of the ensemble.

463 2.7. Economic analysis

464 To judge whether the electricity gains matter economically, we computed
 465 an annual net value per hectare for each method relative to the open field:
 466 electricity revenue minus the annuity and operating cost of the installed ca-
 467 pacity minus the loss of crop revenue (crop production costs are unchanged).
 468 The installed capacity is $2.1 g$ MWp per hectare (modules cover a share g of
 469 the land at 0.21 kWp m^{-2}). Yield is dry matter, valued at assumed prices.
 470 A conventional ground-mounted plant (GCR 0.4) that replaces the crop is
 471 the comparator. All parameters (Table 15) are assumptions chosen to be
 472 representative, not data, and their ranges drive the conclusions; we there-
 473 fore report break-even electricity prices and one-at-a-time sensitivities, and
 474 we do not monetise food security, which market crop prices do not capture.
 475 The analysis omits land rent, module degradation, the extra structure cost of
 476 sparse arrays, curtailment, subsidies and the uncertainty of the parameters
 477 themselves, which the bootstrap intervals do not include.

478 2.8. Evaluation protocol, statistics and upscaling

479 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
 480 ing (70%) or test (30%), so that no test site has a training site in the same
 481 block. Training uses the 2001–2014 seasons of the training sites. Design
 482 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 483 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 484 four climates.

485 We report the following metrics:

- 486 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 487 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 488 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
489 compliance (share of sites whose mean test-season retention is $\geq \rho$);
- 490 • the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- 491 • the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011), which gives
492 equal weight to the retained yield and to the electricity relative to a
493 conventional plant, and is relative to the open field of the same climate
494 (a higher LER under warming reflects the lower reference electricity
495 share and not a benefit of agrivoltaics).

496 Two aggregations are used and labelled in the text: tables report means over
497 all test seasons of a sample (season-pooled), whereas statistical comparisons
498 and intervals use the crop–site pair as the unit of analysis (mean of the pair’s
499 seasons, then mean over pairs), so the same contrast can differ slightly be-
500 tween the two (for example 0.034 and 0.038 for SALS minus the phenology
501 rule on the main split). Samples are also labelled: the main analysis uses
502 the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses
503 pooled over three splits use the 76 distinct pairs of every third held-out pair
504 and a grid with step 0.05, and analyses of alternative shade responses on
505 one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statis-
506 tical comparisons use site means as the unit of analysis. We report paired
507 Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence in-
508 tervals of the mean difference. SALS was trained with three random seeds
509 on the main split, and with one seed on each of two further spatial splits.

510 For the global estimate, every site is evaluated in the split in which it is
511 held out. The mean electricity and yield per hectare of each crop are scaled
512 to the *physical* area of that crop in SPAM 2010, which avoids double counting
513 multiple-cropped land (Waha et al., 2020). We report the installed capacity
514 (TWp), the capacity factor and a supply curve.

515 3. Results

516 3.1. Electricity and food on held-out sites and years

517 Table 2 compares the four controllers on the 92 held-out crop–site pairs of
518 the main split (95 pairs are held out; pairs without a season whose open-field

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

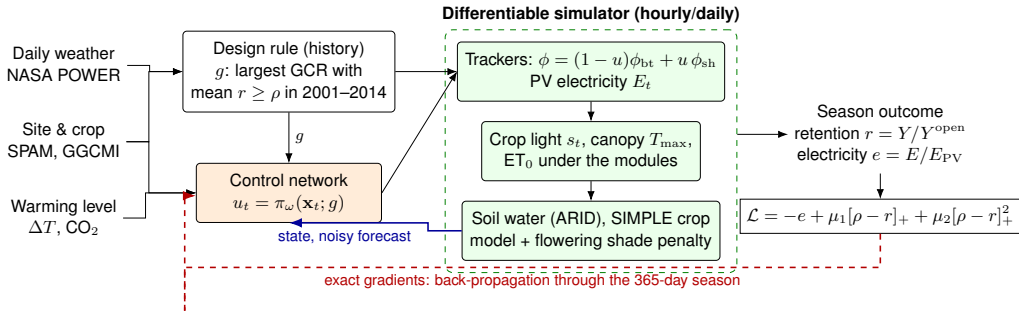


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

yield is at least 0.2 t ha^{-1} in all four climates are excluded) (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule that shares light only while the crop is growing (phenology rule) reaches

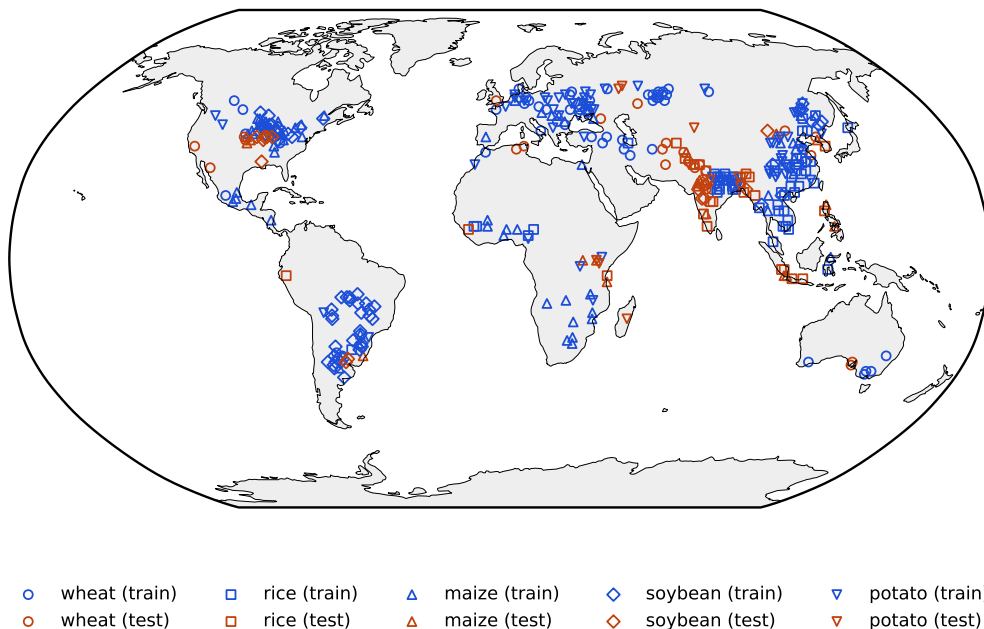


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

531 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the
532 crop on forecast heat or drought days (stress rule) is worse than the phenol-
533 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
534 times when the floor is most binding.

535 **The learned controller is best by a modest, consistent margin.**
536 SALS sustains a density of 17% (similar to the phenology rule), produces
537 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Com-
538 pared with the phenology rule this is +0.038 in LER (95% CI over sites
539 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
540 ble A.18). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
541 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows
542 the same picture at matched densities: the learned policy lies on or slightly
543 above the phenology rule over the whole density range, both lie well above
544 seasonal sharing and the stress rule, and static arrays are far below.

545 **What the floor does and does not guarantee.** Enforcing the floor

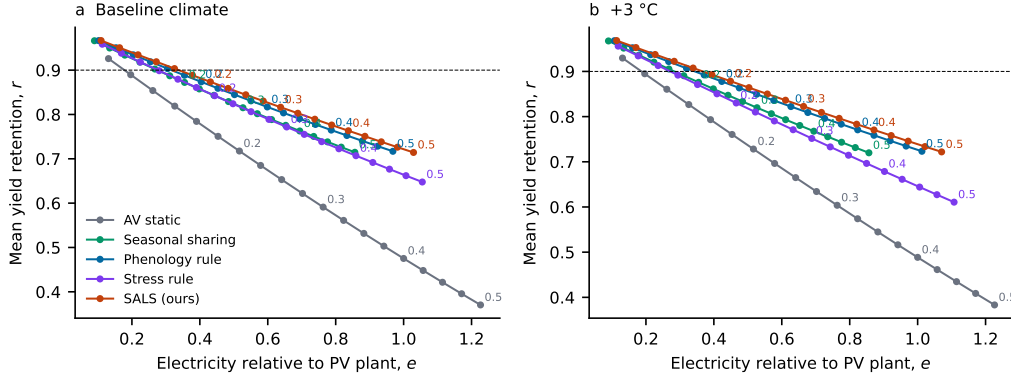


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site's historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-pooled: 0.034 and 0.032; the site-level paired contrasts of Table A.18 are 0.038 and 0.034). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and

568 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
569 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
570 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
571 (0.022, 0.032 and 0.051 at +3 °C).

572 **Which ingredients matter.** Two ablations test the design of SALS
573 at the same training budget. Masking the stress inputs (soil water, ARID,
574 and the maximum temperature, irradiance and ET₀ forecasts) leaves the
575 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
576 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The
577 improvement over the rules therefore does not depend on stress-aware inputs
578 or on training in warmed climates.

579 *3.3. Floor conditioning, chance-constrained training and forecast errors*

580 **One network for all floors.** Regulations and farmers set different
581 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
582 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
583 design step for each floor (Table 3). Its LER exceeds that of the phenology
584 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
585 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without
586 retraining. A network that received the floor as an additional input and was
587 trained over all four floors did not improve on it (differences in LER ≤ 0.001).
588 Because the design step already adapts the density to the floor, conditioning
589 the controller on the floor is not needed.

590 **Layout, rotation limit and harvest-index assumptions.** We checked
591 three assumptions that could drive the ranking (Table 4; 21 pairs of the main
592 split, one training seed). The rotation limit of the trackers sets how much
593 light sharing can deliver: with a limit of 45° the LER of the phenology rule
594 is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at
595 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking
596 is unchanged. Fixed non-tracking arrays held flat or rotated by 30° about
597 the north–south axis (facing east, not the equator; equator-facing tilts are
598 not modelled) do no better than static trackers (LER 1.048 and 1.043 against
599 1.053; differences of 0.005 are within the noise of 21 pairs and one training
600 seed), so static trackers are no worse than the fixed arrays we tested; verti-
601 cal bifacial layouts are not modelled. The ranking of the rules also does not
602 depend on the assumed harvest-index penalty. The window test is weakly in-
603 formative about the timing of the phenology rule, which shares light through
604 almost the whole growing period, so a shifted window cannot move it much;

the informative case is the removal of the penalty. Moving the critical window to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement changes the LER of the phenology rule by at most 0.001, and removing the penalty altogether raises all light-sharing LERs (phenology rule 1.328, SALS 1.353) without changing the order: the phenology rule exceeds seasonal sharing by 0.03 to 0.05 in every case (0.035 with the early window, 0.034 with the late window, 0.053 without penalty), because it produces electricity before canopy development and after maturity, when the crop does not use the light, and not because of the assumed window. The margin of SALS over the written rule changes with these assumptions (0.005 to 0.025 LER), and its margin over the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS equals the tuned rule).

A floor of 66%. The German pre-standard DIN SPEC 91434 requires 66% of the reference yield (DIN, 2021). Repeating the design step with $\rho = 0.66$ (baseline climate, main split, same network trained at 0.9) the ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and SALS 1.71. At this floor the density is limited by the cap and by the mean retention of 0.73 reached at the cap, not by the floor, so the comparison of controllers says little about control; we do not interpret it further.

Chance-constrained training. We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

Forecast errors. We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The stress rule, which depends on the forecast of maximum temperature, degrades with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and its LER from 1.432 to 1.418 as the errors triple.

Tuned rules. The phenology rule above was written, not tuned, whereas

643 SALS is trained. We therefore tuned its window on the training-block seasons
 644 with the same objective as the other rules (electricity at the density where
 645 mean retention equals the floor), first on a coarse 4×3 grid and then, as a
 646 derivative-free baseline, by CMA-ES over a five-parameter schedule (start,
 647 end, two ramp widths and a level; three runs of 200 evaluations, Methods).
 648 The grid-tuned window starts at 0.15 instead of 0.05 of the thermal-time
 649 requirement (Table 5). Two of the three CMA-ES runs, one started at the
 650 grid-tuned window and one at a random point, converged to the same almost
 651 binary window from 0.20 to 0.93 of the requirement at level 1.00; the third
 652 stopped in a worse optimum. Pooled over three splits and 76 distinct crop-
 653 site pairs, the grid-tuned window raises LER over the written rule by 0.026
 654 (95% cluster-bootstrap interval 0.016 to 0.037), the CMA-ES window by
 655 0.037 (0.025 to 0.052) and SALS by 0.044 (0.030 to 0.060), so the optimisation
 656 of a window accounts for about 85% of the margin of SALS over the written
 657 rule. SALS exceeds the CMA-ES window by 0.006 LER (0.001 to 0.012),
 658 which is 1.8% more electricity at equal retention, and the grid-tuned window
 659 by 0.018 (0.012 to 0.024). The remaining margin is small and its interval
 660 only just excludes zero (it covers the pairs only: SALS has one training seed,
 661 and the rules were tuned on training sites of split 0 while the comparison is
 662 pooled over three splits); it is not explained by the tuning of a window or a
 663 ramp, and it may come from the state inputs of SALS (soil water, canopy
 664 development, density), which the rules do not use and which we did not give
 665 them. SALS meets the floor in fewer seasons than the rules (83% against
 666 84–85%).

667 3.4. *What the learned policy does*

668 Averaged over the rainfed test seasons of the main split at a density of
 669 0.3, the light-sharing level of SALS is a smooth function of relative thermal
 670 time. It is close to zero in the first 10% of the thermal time required for
 671 maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and
 672 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau
 673 overlaps the window in which our crop model penalises shading through the
 674 harvest index; we did not test whether this alignment is learned or inherited
 675 from the initialisation. The ramps at both ends let the modules produce
 676 electricity when additional light is worth least to the crop. Fig. 4 shows two
 677 example seasons. The phenology rule, which is also the initialisation of the
 678 network, shares light during the whole growing period. Fine-tuning through
 679 the simulator found that sharing can be reduced at the edges of the season:

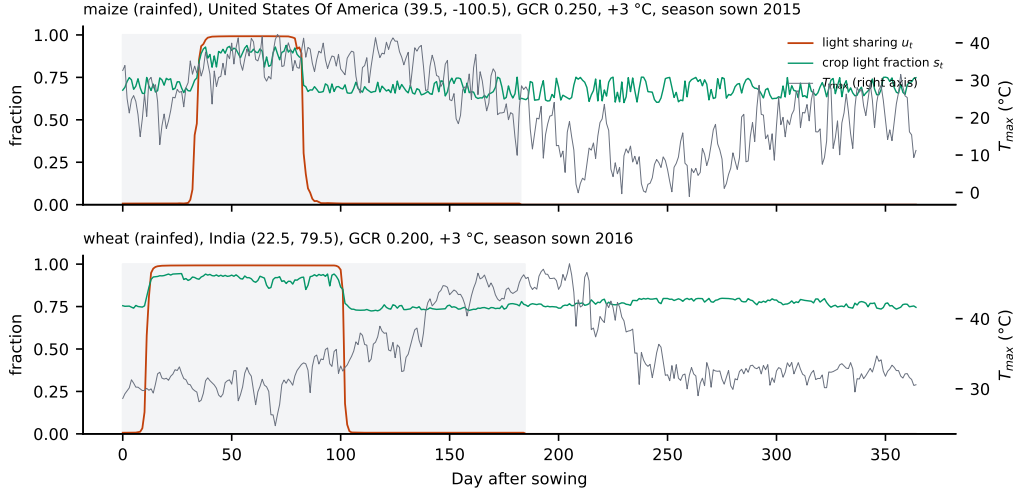


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

680 SALS gains its advantage by yielding light when it matters and collecting
681 electricity when it matters less.

682 The policy also shares less light on days with a forecast heat stress: the
683 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^{\circ}\text{C}$)
684 on other days, and 58–63% of sites show a lower level on heat days. This
685 cannot be read as a stress response, however. Heat days occur late in the
686 season, when the phenological ramp is already falling, and a policy without
687 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
688 The ablation above confirms that stress information does not improve the
689 result.

690 3.5. Crops and regions

691 The benefit of light sharing differs between crops (baseline climate, SALS).
692 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
693 $577 \text{ MWh ha}^{-1} \text{ yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
694 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
695 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
696 1.14–1.21 for the others).

697 The floor is not met evenly. Production-weighted retention is 0.888 for
698 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
699 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
700 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
701 with a production-weighted retention of only 0.862. Africa and the Middle
702 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
703 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
704 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39 pairs
705 without intervals and should be read as indications. A regulatory floor en-
706 forced per site would therefore need a region- and crop-specific design step,
707 not a global one.

708 3.6. Model behaviour and validation

709 **PV output.** The simulated specific yield of an isolated tracker row agrees
710 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
711 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
712 of 5.5% (Section 2.2).

713 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
714 crops under static arrays as a function of the season-mean shading of the crop.
715 Yield declines close to proportionally with shading: 84–89% relative yield at
716 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
717 loses most and potato and maize least. This response is steeper than the *less*
718 *than proportional* losses that Laub et al. (2022) report for small shade levels
719 of C3 cereals and tubers, and it does not reproduce their finding that maize
720 and grain legumes are more shade sensitive than C3 cereals. In the simulator,
721 shading also relieves heat and drought in rainfed maize and soybean, which
722 offsets the higher harvest-index penalty we assigned to them. The absolute
723 effect of shading on each crop is therefore uncertain (Section 4.5).

724 **Comparison with field trials.** We fitted one radiation-use compen-
725 sation coefficient per crop to the yield–shading curves of Laub et al. (2022)
726 using training-block sites only (Methods, calibration script in the repository).
727 The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the
728 proportional model already lies within about 0.02 of the C3-cereal and tu-
729 ber curves, and it cannot lower the maize and legume responses below the
730 proportional case (root-mean-square error 0.16 for maize and 0.08 for soy-
731 bean). Against independent field trials (Table 6) the model underestimates
732 retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and
733 potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al.

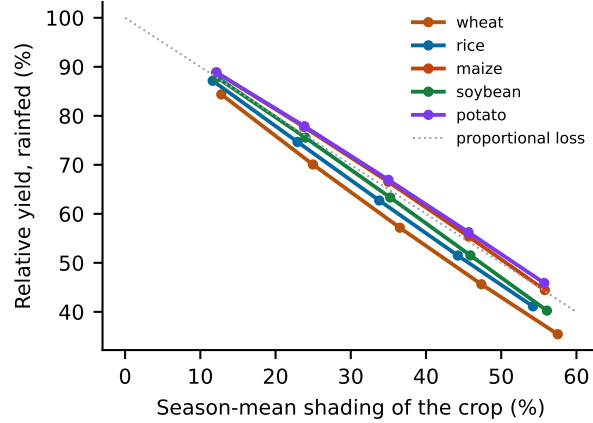


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

(2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 7 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains for rice, wheat and potato are thus conservative with respect to these trials; for maize and soybean the meta-analysis points the other way (Section 3.7).

3.7. A shade-response model learned from field data

Cross-validation by study shows what the data can and cannot support (Tables 8 and 9). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses,

the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22 for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 10). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize and soybean at 40% shading (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index parameters are bounded below at zero. The field trials of maize are also milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6% above the control under a low-density stilt-mounted array in one season at one site. For maize and soybean the direction of the bias is uncertain, and the gains reported in this paper are possibly overestimated.

3.8. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the five crops to the data-driven curves and the field trials (ensemble E2, Table 11). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation

0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the largest compensation (0.91 ± 0.31) and the largest uncertainty; the compensation is an effective parameter that absorbs diffuse-light and light-saturation effects that the simulator does not model explicitly, not a physical value.

Learned controllers. Controllers were evaluated under four fixed responses on 21 crop–site pairs of split 0 with one training seed, so the intervals below, which are cluster bootstraps over pairs of per-pair mean LER differences, do not include seed or split variation (Table 12 gives season means, which differ slightly). Under every response SALS, trained on the proportional response, is better than the written phenology rule: by +0.019 (+0.007 to +0.039), +0.024 (+0.009 to +0.046), +0.038 (+0.018 to +0.062) and +0.015 (+0.001 to +0.037) LER under the proportional, calibrated, field-fitted and harsh responses. It is also above the grid-tuned window and the CMA-ES optimised window, but by less: 0.005 to 0.018 and 0.005 to 0.010 LER, with intervals that exclude zero for the grid-tuned window under 3 of the four responses and for the CMA-ES window under 0. Training on the narrow ensemble E1 did not change this: SALS-robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except for one response) and SALS-adaptive, which sees the measured biomass ratio, by -0.010 to $+0.007$. Training on the wider ensembles did not help either (the robust controllers fall below SALS even under responses inside their training ensemble, which may point to under-training with our budget of 300 steps): SALS-robust on E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017, 0.017 LER under the proportional response, 0.017, 0.021, 0.020 under the calibrated response, 0.004, 0.005, 0.011 under the field-fitted response (intervals contain zero) and 0.008, 0.014, 0.017 under the harsh response; the intervals exclude zero in 7 of the 9 comparisons for the proportional, calibrated and harsh responses. The E3 robust controller is not distinguishable from the written rule (+0.001 to +0.032) under any response (all intervals contain zero). With one seed and 21 pairs these controllers therefore did not retain the margin of SALS within our training budget; a controller that must perform across plausible responses is less specialised, but a larger budget or more seeds could change this. The feedback rule is worse than the written rule by 0.026–0.058 LER under every response (intervals exclude zero). A plausible reason is that the biomass ratio does not reveal the harvest-index loss from shading around flowering, but we did not test this explanation.

Design under uncertainty. The larger risk is in the design step (Tables 13 and 14, Fig. 6). The designs below are evaluated by leaving one

829 ensemble member out as the truth and using the other four, so that the
 830 truth never enters the design. Pooled over the three spatial splits (76 dis-
 831 tinct crop-site pairs), a density chosen under the mean response of E2 leaves
 832 70% of the pairs at the floor when the true response is another plausible
 833 one, against 87% for an oracle design (E3: 71% and 85%). Choosing the
 834 density so that the floor is met in at least 75% of the other members raises
 835 compliance by +7.9 points under E2 (95% cluster-bootstrap interval 4.5 to
 836 11.6) and +7.9 under E3 (3.9 to 12.1), at a cost of 0.031 and 0.025 in LER.
 837 Raising the floor used in the design from 0.90 to 0.92 under the mean re-
 838 sponse does more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and
 839 +14.3 points (9.5 to 19.2) at 0.056 under E3, and the shortfall is smaller
 840 (mean over pairs of the maximum of zero and 0.9 minus retention: 0.0022
 841 against 0.0045 for the quantile rule under E2). The two kinds of rule lie on
 842 one frontier (Fig. 6). Interpolating the margin frontier (floors 0.90 to 0.96)
 843 at the compliance of the ensemble rules, the LER of the margin minus that
 844 of the rule is +0.001 (-0.009 to +0.012) for the rule with 75% of the other
 845 members and +0.003 (-0.016 to +0.021) for the rule with all of them under
 846 E2, and -0.006 (-0.019 to +0.005) and +0.024 (-0.011 to +0.058) under E3
 847 (95% intervals). We find no detectable difference, but the intervals (down
 848 to -0.02 and up to +0.06 LER) are too wide to establish equivalence. The
 849 ensemble is thus useful for sizing the margin (0.02 to 0.04 above the floor for
 850 85–97% of the pairs), not for replacing it. None of these rules is safe out-
 851 side the ensemble (Table 14): under the proportional, calibrated and harsh
 852 responses the mean-response design leaves 7–17% of the pairs at the floor, a
 853 floor of 0.92 or the ensemble rules 12–33%, a floor of 0.94 29–47% and a floor
 854 of 0.96 51–78%, at a cost of about 0.2 LER, whereas the field-fitted response,
 855 which lies within the range spanned by the ensembles, is handled well by
 856 every rule (80–99%) except the mean-response design (68–79%). Outside the
 857 ensemble the shortfall of the pairs below the floor is also large: under E2
 858 the mean shortfall per pair (maximum of zero and 0.9 minus retention) is
 859 0.075, 0.060 and 0.104 for the mean-response design under the proportional,
 860 calibrated and harsh responses, against 0.053, 0.038 and 0.078 for a floor of
 861 0.92 and 0.012, 0.007 and 0.024 for a floor of 0.96. The guarantee therefore
 862 holds only within the range of responses that the ensemble spans; we did
 863 not test an ensemble that spans the proportional and harsh responses, which
 864 would protect against them at a higher cost in LER.

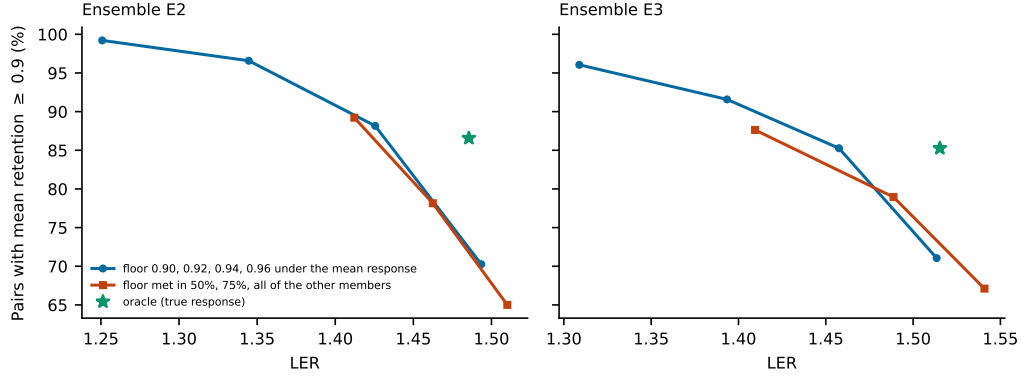


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

865 3.9. Economics of light sharing

866 All figures in this section follow from the assumed parameters of Table 15
867 and should be read as a sensitivity analysis, not a forecast. The economics
868 are dominated by the capacity installed, not by the crop (Table 16): the
869 annuity of the modules is 11–32 kUSD per hectare and year, whereas the
870 crop revenue lost is about 0.3 kUSD, so a method is worth what it earns per
871 installed watt. With the densities designed for the floor, static arrays earn
872 +0.3 kUSD ha⁻¹ yr⁻¹ at 60 USD MWh⁻¹ and break even at 59, whereas all
873 light-sharing methods lose money at that price (−3 to −10 kUSD) and break
874 even at 70 (stress rule), 73 (SALS), 73 (CMA-ES window), 77 (grid-tuned
875 window), 80 (written rule) and 92 (seasonal sharing) USD MWh⁻¹. At 90
876 USD MWh⁻¹ static arrays (6.3 kUSD) are better than the written (4.0) and
877 grid-tuned (5.5) rules, close to the CMA-ES window (7.1) and worse than
878 SALS (7.6), which overtakes static arrays above about 85 USD MWh⁻¹; the
879 difference between SALS and static arrays at 90 USD MWh⁻¹ is +1.3 kUSD
880 (95% interval −0.04 to 2.7, 76 pairs) and not distinguishable from zero,
881 whereas SALS exceeds the CMA-ES window by 0.5 (0.06 to 0.9), the grid-
882 tuned window by 2.1 (1.6 to 2.7) and the written rule by 3.6 (2.8 to 4.5). A
883 conventional plant that replaces the crop breaks even at 48 USD MWh⁻¹ and
884 is more profitable than every agrivoltaic option above that price (16 kUSD
885 at 60, 94 kUSD at 120, against −6 and 21 for SALS); below it all options

lose money and static arrays lose least (at 30 USD MWh⁻¹: plant -23.2, SALS -18.7, static -5.7 kUSD). Per megawatt-hour, the electricity of SALS costs about 73 USD including the crop lost, against 48 USD for the plant, a difference of 25 USD MWh⁻¹. A premium of that size would equalise the unit cost but not the profit per hectare, because a hectare of plant produces three times as much electricity (1306 against 438 MWh): for the same electricity as one hectare of plant, SALS needs about three hectares of array, which costs about 25 USD MWh⁻¹ more including the crop lost but keeps about 90% of the yield of that cropland; relative to the plant, SALS would need a premium of about 49 USD MWh⁻¹ at a market price of 60 and about 109 at 90 to match its profit per hectare, and only about 13 at 60 (none above 73) to match the open field. The sensitivity is large (Table 17), and the ranking of static arrays and light sharing flips with the module cost (at 1.5 USD Wp⁻¹ static arrays are better than SALS, at 0.7 SALS leads by 7 kUSD): at 90 USD MWh⁻¹ the net value of SALS changes from +17 to -8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30-40 USD MWh⁻¹; at 60 14-59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS overtakes them at about 75 (4.9 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), the CMA-ES window at about 79, the stress rule at about 81, the grid-tuned window at about 87 and the written rule at about 97. The value of control is therefore real but depends on a high price: at 90 USD MWh⁻¹ and with profit-optimal densities SALS exceeds the CMA-ES window by 0.4 kUSD ha⁻¹ yr⁻¹ (95% cluster-bootstrap interval 0.1 to 0.8), the grid-tuned window by 1.7 (1.2 to 2.2), the written rule by 2.8 (2.0 to 3.6) and static arrays by 2.0 (0.8 to 3.3); the CMA-ES window exceeds static arrays by 1.6 (0.4 to 2.7), so most of the value of control is obtained with a well-optimised rule.

Cost of the safety margin. At 90 USD MWh⁻¹, moving from a density designed under the mean response to a floor of 0.92 under that response costs 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70% to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points

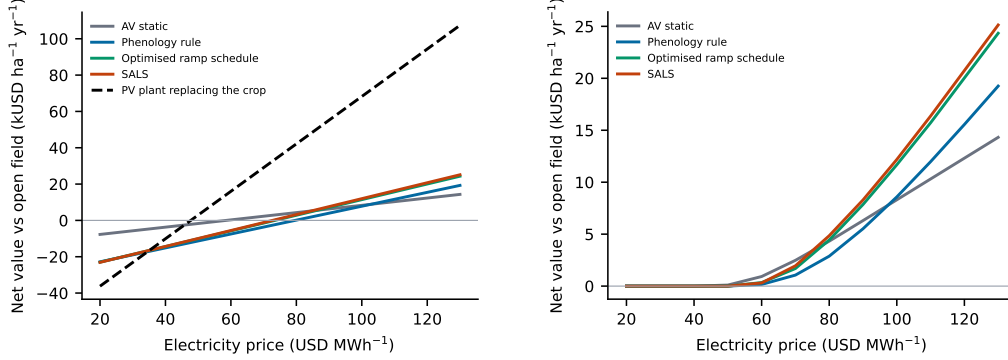


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 15.

(45 and 74 USD per point), so on this measure it is cheaper than the margin under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative cost, because every module beyond break-even loses money; the economics of the design margin therefore depend on the price level, whereas its effect on compliance does not.

3.10. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 8) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize -31%, rice -27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table A.18).

3.11. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split; 222 are held out in at least one split and three have

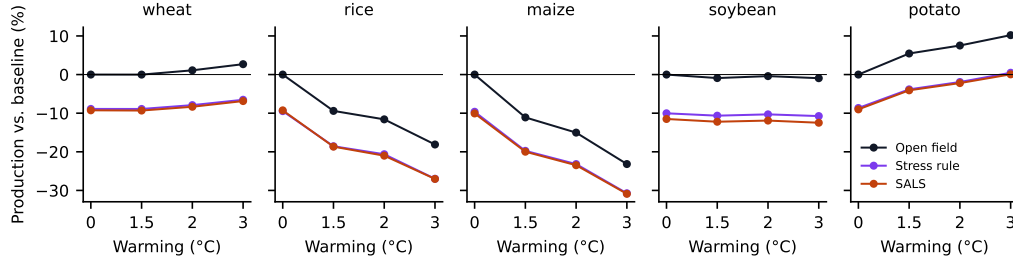


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

no valid season), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{-1} under the baseline climate and 16.0 PWh yr^{-1} at $+3^\circ\text{C}$, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at $+3^\circ\text{C}$), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply 16 PWh yr^{-1} and the best 5% 40 PWh yr^{-1} with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is milder than the meta-analysis (Section 3.7), account for 53% of the electricity of the upscaling (a scaling convention of 5% of the cropland, reported without an interval) (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only wheat, rice and potato would give about 7 PWh yr^{-1} at the

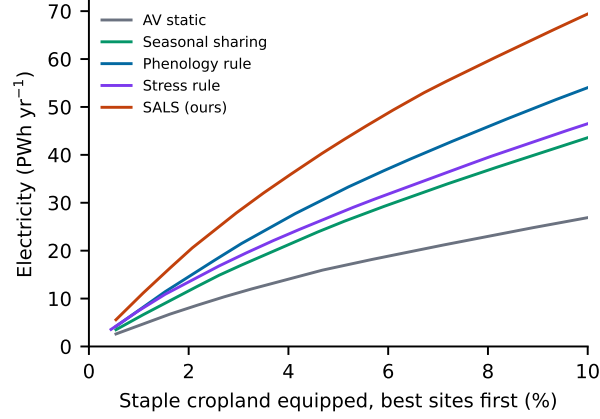


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

970 same coverage, so the global figure should be read as an upper estimate for
 971 about half of its content. By crop, the LER of SALS under the baseline
 972 climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41
 973 for soybean.

974 4. Discussion

975 4.1. What the results establish

976 Four findings stand on consistent evidence. First, sharing light with the
 977 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
 978 can produce under a yield floor. With static trackers the floor limits arrays
 979 to about 6% ground coverage and an LER of 1.07. Any light-sharing con-
 980 trol allows 2.1 to 2.9 times the density, and the LER rises to 1.18–1.24 for
 981 hand-written rules. Second, the best simple rule follows crop phenology. It
 982 shares light from the onset of canopy development (5% of the thermal-time
 983 requirement) until maturity and produces electricity before and after, when
 984 the crop does not use the light. It needs only the crop’s thermal time, which
 985 is observable on the farm, and it is insensitive to forecast errors, whereas
 986 the rule that reacts to heat and drought stress is worse in every climate and
 987 degrades with forecast error. Third, the qualitative result does not depend

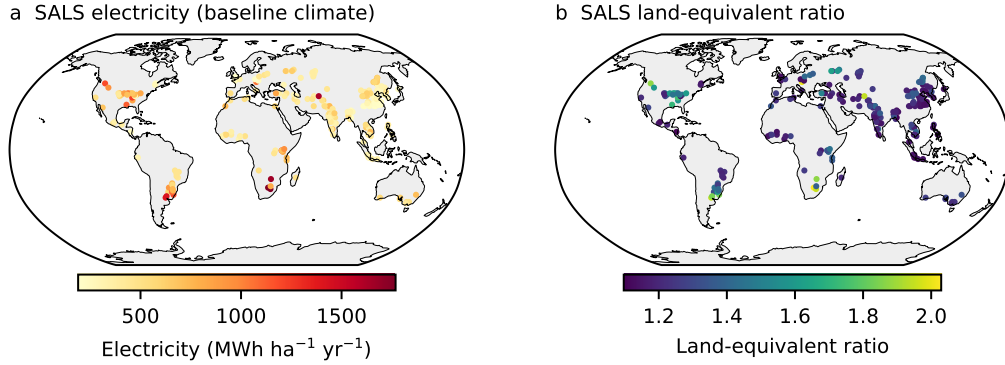


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

on the weakest part of the model. The simulated shade response is conservative against the field trials we could use (Section 3.6); calibrating it raises the allowed density, and the phenology rule stays the best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-fitted responses. However, the response learned from the meta-analysis shows steeper losses for maize and grain legumes than the simulator reproduces (Section 3.7), so the absolute gains are conservative for wheat, rice and potato and uncertain, possibly overestimated, for maize and soybean (53% of the upscaled electricity). Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.044 in LER against the written rule but by only 0.006 (0.001 to 0.012; 1.8% more electricity at equal retention) against a window optimised on the training seasons, reproducibly over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and at 84–86% of the held-out sites. This gain is small compared with the gap between static and dynamic control.

4.2. What role does artificial intelligence play?

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. The learned model is useful mainly as a measurement of how little can be learned: a hierarchical model fitted to 428 observations from 58 studies predicts a held-out study of the four staple crop types with an error of 0.22 in relative yield, no better than proportional loss, and the gain over proportional loss (0.43 to

0.31 over all types) comes from crop types that the simulator does not represent. Crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). Two things follow. The shade response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve, and the process-based simulator does not reproduce the steep losses of maize and legumes seen in the data. Learning the response was limited by the data and not by the model: two studies support the tuber curve, no rice study exists, and 46 panel observations are all leafy vegetables.

The second is calibration through the differentiable simulator. Gradient descent through whole seasons fitted the response of each crop to the learned curves and to the field trials, and it produced an ensemble of plausible responses that fits the trials and the curves for cereals and potato but not the maize and soybean curves (Section 3.8). The ensemble informs the design decision: a density chosen under the mean response leaves about 70% of crop–site pairs at the floor if another plausible response holds, against 85–87% for an oracle, and a density chosen to meet the floor in 75% of plausible responses recovers about half of that gap at a cost of 0.03 LER. A simple margin, however, does as well: raising the floor from 0.90 to 0.92 under the mean response reaches 85–88% of the pairs, and interpolating the margin frontier at the compliance of the ensemble rules gives LER differences (margin minus rule) of +0.001 (−0.009 to 0.012) and +0.003 (−0.016 to 0.021) under E2 and −0.006 (−0.019 to 0.005) and +0.024 (−0.011 to 0.058) under E3, so the two are not distinguishable on this sample, and the cost per point of compliance is similar. None of the rules is safe outside the ensemble (7–33% of the pairs at the floor under the proportional, calibrated and harsh responses for the mean design, the margin of 0.92 and the ensemble rules), and only a floor of 0.96 reaches 51–78% at a cost of about 0.2 LER. What the ensemble adds is the size of the margin, not a better rule. The quantile rule is a scenario-type design (Calafiore and Campi, 2006; Campi and Garatti, 2008); with five members it has none of the finite-sample guarantees of that theory, and the comparison with the margin shows that it is not needed in practice. The bottleneck of tracker-based agrivoltaics under a yield-retention floor is the uncertainty of the crop response that enters the design, and it is data on that response, not larger networks, that would improve the result; the learned curves show a spread of about 20 percentage points between studies. The guarantee holds only within the range of responses that the ensemble

1049 spans.

1050 The third is control, and here learning contributes little. Differentiability is what lets a network of 5441 weights be trained in 300 gradient steps
1051 through whole seasons; a derivative-free search such as CMA-ES handled a
1052 five-parameter schedule with 200 simulator evaluations per run and found
1053 an almost binary window that is only 0.006 LER behind SALS, but would
1054 not scale to the network. The learned controller adds 0.006 LER over an
1055 optimised window (and 0.018 over a grid-tuned one, 0.044 over the written
1056 rule), and 0.005–0.010 over the optimised window under the four alternative
1057 responses on one split (21 pairs, intervals containing zero); a reader who
1058 needs a simple, auditable controller can use that window and obtain about
1059 85% of the electricity gain of SALS over the written rule. Neither of the two
1060 extensions that we expected to matter under uncertainty did: training on
1061 ensembles of responses (E1, E2 and E3) did not help and, for the wider en-
1062 sembles, did not retain the margin over the written rule within our training
1063 budget (one seed, 21 pairs), and neither did giving the controller a farm-
1064 measurable signal, the biomass of the crop relative to a reference plot, nor
1065 a feedback rule built on that signal. The learned policy also has to be ini-
1066 tialised from the phenology rule (Section 4.5). We therefore do not claim that
1067 learned control is what makes agrivoltaics work: in this model, the schedule
1068 that a learned policy converges to is a smooth function of crop phenology,
1069 which a rule can already express.
1070

1071 *4.3. What the results do not establish*

1072 The early hypothesis that stress-aware control, that is, shading the crop
1073 on heat and drought days, would be a main source of benefit under warming
1074 was not supported. The stress rule performs worse than the phenology rule
1075 in every climate, and a learned controller without stress inputs performs as
1076 well as one with them. A plausible reason is that in our crop model the yield
1077 gain from shading on a hot or dry day is smaller than the loss of light on
1078 that day, so stress information does not help. Whether this holds for crop
1079 models with more detailed canopy-temperature and heat-stress responses is
1080 an open question. Agrivoltaics does not protect staple production against
1081 warming in our simulations: light-sharing systems produce about 90% of the
1082 open-field yield in every climate and follow the same decline in maize and
1083 rice (Fig. 8).

4.4. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.11) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

Who bears the risk. A floor defined on the multi-year mean leaves the farmer exposed in the seasons below it: with the density designed for the floor, about 30% of seasons fall short on the main split (70% compliance for SALS), the shortfall is mostly small (2.7% of SALS seasons are below 0.8 of the open-field yield and the 10th percentile of retention is 0.883), and a per-season guarantee through chance-constrained design costs 0.065 LER. Siting by electricity favours maize and soybean areas, where compliance is lowest (Section 3.11), so the risk concentrates where the energy gain is largest. The crop revenue at stake (about 0.3 kUSD ha⁻¹ yr⁻¹) is small against 20–30 kUSD of electricity revenue, so the exposure is mainly one of compliance with the floor and of the contract, and it coincides with the largest uncertainty of the shade response, that of maize and soybean. Who carries it, through contracts, insurance or a premium, is a design question that the model does not address, and food is valued only through crop revenue at market prices.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias −4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the field trials of rice, wheat and potato; for maize it is conservative with respect to the trials but milder than the meta-analysis, and likewise for soybean, where the 40%-shading relative yields are 0.42 and 0.40 in the data against 0.75 and 0.48 in the simulator, so for these two crops the direction of

the bias depends on the source. The harvest-index sensitivities h are assumptions, and the sensitivity analyses use subsamples of crop-site pairs (22 for the rule comparison under alternative responses, 21 for the controller comparison, one training seed).

- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency, and its CO₂ response is that of controlled experiments that tend to overstate field responses (Long et al., 2006). More detailed models with grain-number formation and canopy energy balance may favour different schedules.
- *Layouts and models.* The baseline of static arrays is a backtracking tracker held at the energy-optimal angle; flat and 30°-rotated (east-facing, not equator-facing) fixed arrays did no better in a check on 21 pairs, but vertical bifacial layouts (Willockx et al., 2023; Campana et al., 2021) are not modelled, and the ground coverage ratio is a coarse proxy of crop productivity (Dupraz, 2023). The PV model is isotropic in the diffuse component and has no bifacial gain; it was benchmarked against PVGIS at low density only and has not been compared with a reference implementation such as pvlib (F. Holmgren et al., 2018). The rotation limit was varied between 45 and 80 degrees and changes the size of the benefit but not the ranking; the harvest-index window is assumed to be common to all crops, and shifting it or removing the penalty did not change the ranking of the rules.
- *Climate scenarios.* Warming was applied as uniform temperature shifts with fixed precipitation, radiation and relative humidity, and with unchanged cultivars and sowing dates. It is not a bias-adjusted climate projection.
- *Initialisation and state information.* The network is initialised from the phenology rule, because training from random weights collapsed to full

sharing. The gain of SALS is therefore the gain of fine-tuning that rule, and we cannot say whether a better optimiser or a richer policy class would find larger gains. The controller uses the simulated state of the crop and soil (thermal time, canopy interception, soil water) without measurement error, and the forecast errors we tested apply to weather only.

- *Economics and deployment.* The economic analysis (Section 3.9) uses assumed prices and costs, no grid, curtailment, storage or subsidy model, and does not value food security; its break-even prices move with the capex and the discount rate (Fig. 7). The value of midday electricity, curtailment, storage, structural cost and farmer acceptance (Trommsdorff et al., 2021) will determine the realisable share. The average row pitch at the chosen densities (about 13 m for 2.2-m modules) leaves room for machinery, but the densest designs (GCR up to 0.5) have a pitch of only 4.4 m.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever on the electricity of tracker-based agrivoltaics at a given yield-retention floor. In a differentiable simulator, evaluated on held-out crop–site pairs (92 on the main split, 76 distinct pairs pooled over three splits for the controller and design comparisons) with the density of each site chosen from its own history, light-sharing rules allowed 2.1 to 2.9 times the module density of static arrays at a mean yield retention of 90% and raised the land-equivalent ratio from 1.07 to 1.18–1.24 under the proportional-growth shade response. A rule that follows crop phenology and needs no weather forecast was the best written rule. The lever pays only at high electricity prices: with illustrative costs it breaks even with the open field at 70 to 92 USD MWh^{−1} (73 for the learned controller, 59 for static arrays), static arrays earn more below about 75 to 85 USD MWh^{−1}, and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh^{−1}. In this model agrivoltaics therefore pays only with a premium, land-use rules or a value for food production that the model does not include.

The crop’s response to shade, the least certain input, was learned from 428 field observations. For the staple crop types a hierarchical model predicted a

held-out study no better than proportional loss (error 0.22), which measures the between-study spread, and gradient descent through the simulator turned the learned curves and four field trials into ensembles of plausible responses. The simulator is conservative against the trials for rice, wheat and potato; for maize and soybean, which supply 53% of the upscaled electricity, the meta-analysis indicates larger losses than simulated and one maize trial smaller ones, so the gains for these crops are uncertain and possibly overestimated. A learned controller exceeded the written phenology rule by 0.044 LER but a CMA-ES-optimised window by only 0.006 (95% interval 0.001–0.012, one training seed). The source of this margin is not identified: it is not explained by the tuning of a window or a ramp, and we did not give the rules the soil-water and density inputs that the controller has. Training on ensembles of shade responses, or giving the controller a farm-measurable biomass signal, did not help within our budget (one seed, 21 pairs).

The design step is the larger source of compliance risk. A density chosen under the mean response left about 70% of crop-site pairs at the floor when another plausible response held, against 85–87% for an oracle design; this gap of 15 points exceeds the 4 points between light-sharing controllers at a fixed design rule. Designing for the floor in 75% of the plausible responses recovered about half of the gap at a cost of 0.03 LER, and a floor raised by 0.02 under the mean response gave no detectable difference at equal compliance (the intervals include zero and are too wide to establish equivalence); no rule was safe outside the ensemble. The ensemble is therefore useful to size a safety margin, not to replace it. The yield floor is met on average but not in every season or region, and the electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹, a scaling convention without an interval) requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with more field data, notably for maize and legumes where the meta-analysis and the trials disagree, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the yield-retention floor per site and per season.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

1226 Declaration of competing interest

1227 The author has no known competing financial interests or personal re-
1228 lationships that could have appeared to influence the work reported in this
1229 paper.

1230 Data and code availability

1231 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
1232 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
1233 POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao
1234 et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022)
1235 (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository).
1236 The code of the simulator, of SALS and of all experiments, together with the
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1238

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1242 Appendix A. Statistical comparison

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

Setting	Static tracker	Phenology rule	Tuned rule	SALS
Main (limit 60°, window 0.45–0.65)	1.053	1.170	1.180	1.189
Rotation limit 45°	1.057	1.138	1.144	1.144
Rotation limit 80°	1.053	1.206	1.219	1.224
Window 0.30–0.50	1.053	1.169	1.176	1.185
Window 0.60–0.80	1.053	1.171	1.181	1.182
No harvest-index penalty	1.109	1.328	1.347	1.353

Table 5: Rules and the learned controller pooled over the three spatial splits (76 distinct crop–site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The grid-tuned window shares light while thermal time lies between 0.15 and 1.0 of the season requirement (best of a 4×3 grid on training-block seasons of split 0; the written rule uses 0.05 to 1.0). The CMA-ES schedule has five parameters (start, end, two ramp widths, level) and converged to an almost binary window from 0.20 to 0.93 of the requirement at level 1.00 (two of three runs; see Methods).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule (written)	0.154	0.292	0.916	85	91	1.207
Grid-tuned window	0.161	0.317	0.916	84	89	1.233
CMA-ES optimised window	0.160	0.329	0.916	83	87	1.245
SALS	0.161	0.335	0.916	83	87	1.251

Table 6: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 7: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 8: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 9: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 10: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 11: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 12: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule (written)	1.164	1.190	1.456	1.122
Grid-tuned window	1.170	1.199	1.472	1.129
CMA-ES optimised window	1.173	1.198	1.479	1.125
SALS	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 13: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop-site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair.

Design	E2		LER	E3		
	Pairs ≥ 0.9 (%)	Shortfall		Pairs ≥ 0.9 (%)	Shortfall	
oracle (true response)	87	0.0025	1.486	85	0.0029	1
mean response	70	0.0062	1.493	71	0.0093	1
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1
floor met in all other members	89	0.0019	1.412	88	0.0032	1
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1

Table 14: Design rules outside the ensemble: share of crop-site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). Designs use the whole ensemble E2 or E3 (five members and their mean); the oracle uses the true response. LER in the last two columns is the mean over the four responses; the oracle rows are identical for E2 and E3 because the oracle uses the true response.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 15: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 16: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 15). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity + O&M	Crop loss	Net at 60	Net at 90
AV static	200	11.5	0.27	0.3	6.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5
Stress rule	336	23.2	0.30	-3.3	6.8
Phenology rule (written)	382	30.1	0.30	-7.5	4.0
Grid-tuned window	415	31.6	0.31	-7.0	5.5
CMA-ES optimised window	430	31.3	0.31	-5.8	7.1
SALS	438	31.6	0.31	-5.6	7.6
Conventional PV plant	1306	58.7	3.60	16.0	55.2

Table 17: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

Parameter	Value	Static arrays	Optimised window	SALS	Conventional plant
Base		6.3	7.1	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	16.5	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-8.5	-8.2	25.8
Discount rate	4%	8.0	11.9	12.4	64.1
Discount rate	8%	4.4	1.9	2.4	45.4
Crop prices	0.5×	6.4	7.3	7.7	57.0
Crop prices	2.0×	6.0	6.8	7.3	51.6
Electricity price	60	0.3	-5.8	-5.6	16.0
Electricity price	120	12.3	20.0	20.7	94.4

Table A.18: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). Δ LER: mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	Δ LER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92